Exact equations

Exact differential equations are another form of first order differential equation for which we can find a solution. In order for a differential equation to be called an **exact differential equation**, it must be given in the form

$$M(x, y) + N(x, y)\frac{dy}{dx} = 0$$

This kind of equation will have an implicit solution $\Psi(x, y) = c$ (Ψ is the Greek letter Psi ("sigh")) defined in two variables x and y, where c is a constant and where Ψ has continuous partial derivatives such that

$$d\Psi(x, y) = M(x, y)\ dx + N(x, y)\ dy$$

Test for an exact equation

If we say that the functions $M(x, y)$ and $N(x, y)$ have continuous partial derivatives, then the differential equation

$$M(x, y) + N(x, y)\frac{dy}{dx} = 0$$

is exact if and only if $M_y = N_x$, or

$$\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x}$$

If we put this in words, we can say

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> If the partial derivative of M with respect to y is equal to the partial derivative of N with respect to x, then the differential equation is exact.

This is a test we can use to say whether or not the differential equation is exact, before we go about finding the solution $\Psi(x, y)$.

Solution to an exact equation

The implicit solution to an exact differential equation is given by

$$\Psi(x, y) = c$$

where c is a constant. The partial derivatives of Ψ with respect to x and y are

$$\frac{\partial \Psi}{\partial x} = \Psi_x = M(x, y) = M$$

$$\frac{\partial \Psi}{\partial y} = \Psi_y = N(x, y) = N$$

The left sides of these equations are partial derivatives of Ψ, but we need to get back to Ψ itself. To do that, we'd need to take the integrals

$$\int \Psi_x \, dx = \int M(x, y) \, dx$$

$$\int \Psi_y \, dy = \int N(x, y) \, dy$$

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Taking the integral with respect to x of the partial derivative with respect to x cancels both operations and leaves us with just Ψ, in the same way that taking the integral with respect to y of the partial derivative with respect to y cancels both operations and leaves us with just Ψ.

$$\Psi = \int M(x, y)\ dx + h(y)$$

$$\Psi = \int N(x, y)\ dy + h(x)$$

In other words, if we want to find an equation for Ψ, we can either take the integral of M with respect to x, or the integral of N with respect to y. Both integrals will work, so we should look at M and N and then choose whichever function looks easier to integrate.

Remember that when we use the first integral (the one with M), we're integrating a multivariable function in terms of x and y with respect to x only. Which means that, instead of adding C to account for the constant of integration after we integrate, we have to add $h(y)$ to account for a function in terms of y.

Similarly, when we use the second integral (the one with N), we're integrating a multivariable function in terms of x and y with respect to y only. Which means that, instead of adding C to account for the constant of integration after we integrate, we have to add $h(x)$ to account for a function in terms of x.

Then it's just a matter of solving for $h(y)$ or $h(x)$, which we'll do by differentiating Ψ with respect to y if we're trying to find $h(y)$, or with respect to x if we're trying to find $h(x)$.

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That differentiation process will give us either Ψ_x with $h'(x)$ or Ψ_y with $h'(y)$. We also know that $M(x,y) = \Psi_x$ or $N(x,y) = \Psi_y$, we'll make that substitution and then simplify the equation to solve for $h'(y)$ or $h'(x)$. Then we can integrate both sides of the remaining equation to solve for $h(y)$ or $h(x)$.

Finally, we'll plug $h(y)$ or $h(x)$ back into the equation for Ψ, set the equation equal to c, and this will be the implicit solution to the exact differential equation.

In summary, to find the solution of an exact differential equation, we'll

1. Verify that $M_y = N_x$ to confirm the differential equation is exact.
2. Use $\Psi = \int M(x,y)\ dx + h(y)$ or $\Psi = \int N(x,y)\ dy + h(x)$ to find $\Psi(x,y)$, including a value for $h(y)$ or $h(x)$.
3. Find $\dfrac{\partial \Psi}{\partial y} = \dfrac{\partial}{\partial y}\left(\int M(x,y)\ dx\right) + h'(y) = N(x,y)$, or find $\dfrac{\partial \Psi}{\partial x} = \dfrac{\partial}{\partial x}\left(\int N(x,y)\ dy\right) + h'(x) = M(x,y)$.
4. Integrate both sides of the equation to solve for $h(y)$ or $h(x)$.
5. Plug $h(y)$ or $h(x)$ back into Ψ in step 2.
6. Set $\Psi(x,y) = c$ to get the implicit solution.

Let's try an example where we find the general (implicit) solution to an exact differential equation.

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Example

If the differential equation is exact, find its solution.

$$x^3y^4 + (x^4y^3 + 2y)\frac{dy}{dx} = 0$$

First, we'll test to see whether or not the differential equation is exact. Matching the given equation to the standard form of an exact differential equation, we can say that

$$M(x, y) = x^3y^4$$

$$N(x, y) = x^4y^3 + 2y$$

We'll test to see whether $M_y = N_x$ (Step 1).

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial}{\partial y}(x^3y^4) = \frac{\partial}{\partial x}(x^4y^3 + 2y)$$

$$4x^3y^3 = 4x^3y^3$$

Since $M_y = N_x$, we know that the differential equation is exact, so now we just need to find the solution $\Psi(x, y) = c$.

The functions $M(x, y)$ and $N(x, y)$ are equally easy to integrate, so we'll just use $M(x, y)$, and then Ψ can be given by

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$$\Psi = \int M(x,y)\ dx \text{ (Step 2)}$$

$$\Psi = \int x^3y^4\ dx$$

$$\Psi = \frac{1}{4}x^4y^4 + h(y)$$

We had to add $h(y)$ instead of just C because we integrated a multivariable function with respect to x only, which doesn't account for the integration of y. Now we need to find $h(y)$, which we'll do by differentiating both sides with respect to y.

$$\Psi_y = x^4y^3 + h'(y) \text{ (Step 3)}$$

Because we know that $\Psi_y = N(x,y)$, we'll make that substitution and then solve for $h'(y)$.

$$x^4y^3 + 2y = x^4y^3 + h'(y)$$

$$2y = h'(y)$$

To find $h(y)$, we'll integrate both sides of this equation with respect to y.

$$\int 2y\ dy = \int h'(y)\ dy \text{ (Step 4)}$$

$$y^2 + k_1 = h(y) + k_2$$

$$h(y) = y^2 + k_1 - k_2$$

$$h(y) = y^2 + k$$

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Plugging this value for $h(y)$ into the equation for Ψ in step 2 gives

$$\Psi = \frac{1}{4}x^4y^4 + h(y) \text{ (Step 5)}$$

$$\Psi = \frac{1}{4}x^4y^4 + y^2 + k$$

Finally, setting $\Psi = c$ to find the solution to the exact differential equation, we get

$$c = \frac{1}{4}x^4y^4 + y^2 + k \text{ (Step 6)}$$

$$c - k = \frac{1}{4}x^4y^4 + y^2$$

$$c = \frac{1}{4}x^4y^4 + y^2$$

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